

A Robust Display Delay Compensation Technique Considering User's Head Motion Direction for Cloud XR

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ABSTRACT

Conventionally, it has been difficult to realize both photorealistic and geometrically consistent 3DCG rendering on head mounted displays (HMDs), due to the trade-off between rendering quality and low motion-to-photon latency (M2PL). In order to solve this problem, we propose a novel rendering framework, where the server renders RGB-D images of a 3D model with the optimally arranged rendering viewpoints, and the client HMD realizes geometric compensation of M2PL by employing depth image based rendering (DIBR). Experiments with real smart glasses show that the proposed method can display binocular images closer to the ground truth than the conventional approaches.

CCS CONCEPTS

• Computing methodologies → Computer graphics;

KEYWORDS

cloud xr, display delay compensation, depth image based rendering

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1 INTRODUCTION

A cloud-based 3DCG rendering technique that utilizes high-capacity and low-latency mobile networks such as 5G has been proposed, and has already been utilized in some cloud XR (cross reality) services. Although the transmission latency is reduced by using multi-access edge computing (MEC), etc., the motion-to-photon latency (M2PL) increases because of the network and codec delays compared to standalone XR applications. Then, XR services targeting head mounted displays (HMDs) face the problem of cybersickness and degradation of the user experience. As an approach to compensate for the M2PL, a 6-DoF head motion prediction method has been considered [Gül et al. 2020]. However, it cannot follow the user's sudden or irregular head motion and has not yet solved the problem.

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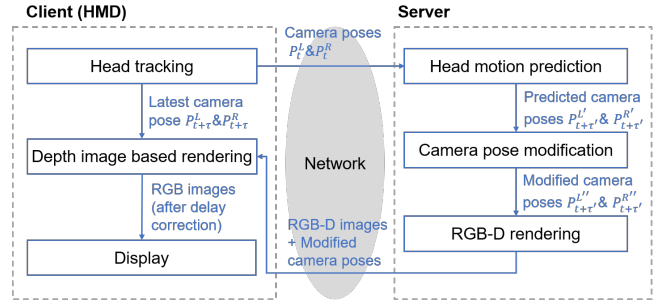


Figure 1: Proposed cloud XR rendering framework

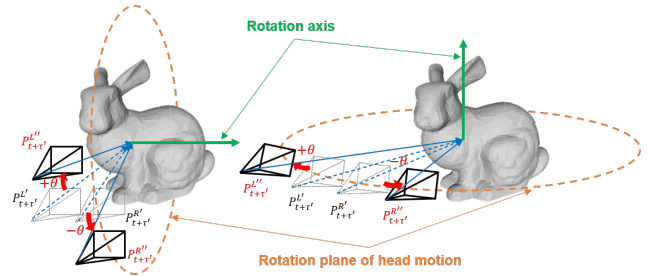


Figure 2: Proposed camera pose modification (CPM)

As an approach to fundamentally solve this problem, depth image based rendering (DIBR) [Mori et al. 2008] is being employed for cloud rendering. DIBR can convert the delayed images into the images consistent with the latest head poses by using RGB-D images. Although several methods that perform DIBR on an HMD have already been proposed, they still have some problems, such as a large amount of information (e.g., multi-view RGB-D images more than two viewpoints) required to be transmitted to suppress disocclusion (causing network load and further M2PL increase) [Reinert et al. 2016] and the lack of evaluation in the actual 6DoF head poses.

In this paper, we propose a method to overcome the problem of the disocclusion with only two RGB-D images for binocular viewing. Our method utilizes the results of the 6DoF head pose prediction and adjusts the viewpoints for 3DCG rendering considering the user's head motion direction. Through the experiments using head pose data acquired with actual smart glasses, the effectiveness by the proposed scheme is shown compared to the conventional methods.

2 PROPOSED RENDERING FRAMEWORK

Our proposed rendering framework is shown in Figure 1. First, as in the conventional prediction-based cloud rendering approach, the server performs the head motion prediction to compensate for the M2PL using the camera poses received from the HMD. As a result,

the future camera poses after assumed delay time τ' is predicted. For the prediction, we apply an autoregressive model in which the translational and rotational components of the 6DoF camera pose are learned separately [Gül et al. 2020].

The error in the head motion prediction becomes larger in proportion to M2PL, and the discrepancy between the predicted image and the ground truth tends to increase. DIBR can compensate for this discrepancy, but if the prediction error is large, the quality of the corrected image degrades due to the disocclusion problems such as enlargement of the missing area. Large prediction errors are caused by sudden acceleration or slowdown of head motion, and tend to occur in the progressive or regressive direction of the head motion. Therefore, the proposed method modifies the rendering viewpoints of binocular images according to the predicted head motion direction. Figure 2 shows the image of the proposed camera pose modification (CPM). Here, the rotation axis is set to pass through the model center and be orthogonal to the rotation plane of head motion, and the rotation plane is defined as a plane passing through the three points: the model center, the central viewpoint of camera poses P_t^L and P_t^R , and the central viewpoint of the predicted camera poses. In CPM, the two predicted camera poses are rotated around the rotation axis by a rotation angle θ (5 degrees was adopted as the best accuracy in preliminary experiments) in the progressive and regressive directions, respectively, to cover the model surface where disocclusion is likely to occur.

After rendering 3DCG with the modified camera poses, RGB-D images are acquired from the rendering buffers and transmitted to the client together with the modified camera poses. At the HMD, we apply DIBR to the received binocular RGB-D images using the latest camera poses, and perform view synthesis to correct the head motion prediction error.

3 EXPERIMENTAL RESULTS

For 12 subjects, viewing experiment through NrealLight smart glasses was conducted. In the experiment, a static 3D model of a full-sized person was presented, and 6DoF pose sequences of both eyes with time information was recorded. From each of the recorded sequences, we cropped out the segments that contained relatively large head movements (about 10 seconds for each person) and used them as the dataset for evaluation. The 3D models for evaluation were rendered with the pose sequences, and the RGB-D test sequences (640×480 pix.) were created by adding a predetermined delay (0–300 msec.). We used two types of 3D models (Bunny and Dragon [McGuire 2017]) for the evaluation. The test images with a delay of 0 were treated as the ground truth. Then, we compared the head motion prediction method (Pred [Gül et al. 2020]) based on the autoregressive model ($\rho = 32$, sampling interval 5 msec.), the DIBR method using the binocular RGB-D images (DIBR [Mori et al. 2008]), and the method of using DIBR and CPM together (Proposed).

In Figure 3, the results of PSNR and MSSIM are shown for the proposed scheme, Pred, and DIBR, respectively. The horizontal axis indicates M2PL. The performance of Pred degrades rapidly as M2PL increases, due to the frequent head motion. On the other hand, DIBR greatly improves the robustness to M2PL, and the performance degradation is slow as M2PL increases. Furthermore, applying the proposed CPM further improves the PSNR of DIBR

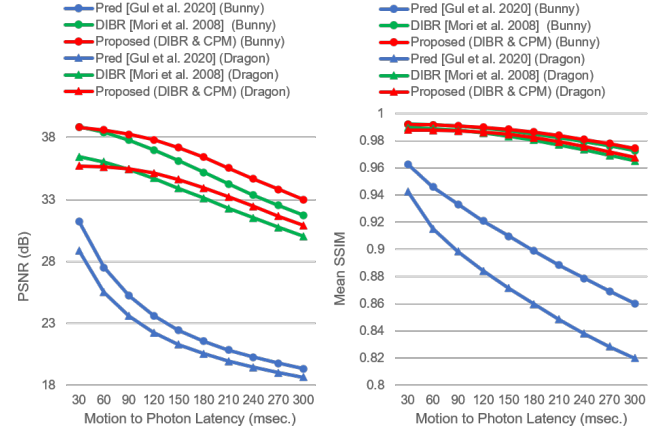


Figure 3: M2PL correction accuracy

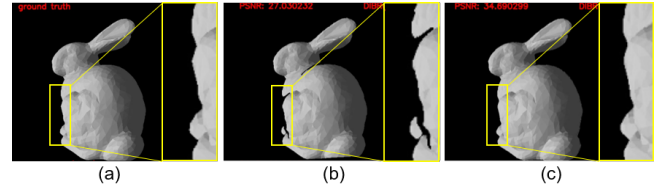


Figure 4: Examples of displayed images when M2PL is 210msec: (a) ground truth, (b) DIBR, (c) proposed method. In this scene, the proposed method improves the PSNR by 7.0 dB and MSSIM by 0.082.

when M2PL is larger than about 90 msec. Compared to the Bunny model, the rendering performance of the Dragon model is generally worse. This is due to the fact that the Dragon model has a more complex shape and is more prone to the disocclusion problems. When compared with MSSIM, the average improvement of the proposed method on DIBR is relatively smaller than that of PSNR. However, Figure 4 shows that the improvement in image quality by the proposed method is visually noticeable in the case where the proposed method eliminates the disocclusion of DIBR.

4 CONCLUSIONS

In this paper, we proposed a novel rendering framework that can display photorealistic 3DCG on smart glasses with high geometric consistency. Through the experiments using actual head pose sequences, we have shown that the proposed method is more robust to M2PL than the conventional motion prediction-based approach and the DIBR method. In the future, we plan to verify the effectiveness of the proposed method by subjective evaluation experiments.

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